

Luca Buiarelli

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Profile

Material Science PhD candidate with a physics background, specializing in first-principles theory of magnetic multipolar order, symmetry-constrained response functions, and unconventional magnetism in real materials, with experience connecting group-theoretical predictions to experimental observables.

Education

University of Minnesota, USA — PhD, Materials Science	<i>Sep. 2023 – Present</i>
University of Copenhagen, Denmark — MSc, Physics	<i>Sep. 2021 – May 2023</i>
University of Pisa, Italy — BSc, Physics	<i>Sep. 2018 – Sep. 2021</i>

Articles

Published

- Interplay between charge correlations and superconductivity across the superconducting domes of $\text{CsV}_3\text{Sb}_{5-x}\text{Sn}_x$
Andrea N. Capa Salinas, Brenden R. Ortiz, Steven J. Gomez Alvarado, Sarah Schwarz, Ganesh Pokharel, **Luca Buiarelli**, Hyeonseo Harry Park, Shiyu Yuan, Roland Yin, Suchismita Sarker, Turan Birol, Stephen D. Wilson
Phys. Rev. B **114** (2026) [Journal] [arXiv]
- Probing multipolar order in the candidate altermagnet MnF_2 through the elastocaloric effect under strain
Rahel Ohlendorf, **Luca Buiarelli**, Hilary ML Noad, Andrew P Mackenzie, Rafael M Fernandes, Turan Birol, Jörg Schmalian, Elena Gati
Phys. Rev. Lett. **137** (2026) *Editors' Suggestion* [Journal] [arXiv]
- Observation of mirror-odd and mirror-even spin texture in ultra-thin epitaxially-strained RuO_2 films
Yichen Zhang, Seung Gyo Jeong, **Luca Buiarelli**, Seungjun Lee, Yucheng Guo, Jiaqin Wen, Hang Li, Sreejith Nair, In Hyeok Choi, Zheng Ren, Ziqin Yue, Alexei Fedorov, Sung-Kwan Mo, Junichiro Kono, Jong Seok Lee, Tony Low, Turan Birol, Rafael M. Fernandes, Milan Radovic, Bharat Jalan, Ming Yi
Science Adv. **12** (2026) [Journal] [arXiv]
- Electronic and structural properties of Rh- and Pd-based kagome layered shandites from first principles
Luca Buiarelli, Turan Birol, Brian M. Andersen, Morten H. Christensen
Phys. Rev. B **113** (2026) *Editors' Suggestion* [Journal] [arXiv]
- Altermagnetic Polar Metallic Phase in Ultra-Thin Epitaxially-Strained RuO_2 Films
Seung Gyo Jeong, In Hyeok Choi, Sreejith Nair, **Luca Buiarelli**, Bitu Pourbahari, Jinyoung Oh, Nabil Bassim, Ambrose Seo, Woo Seok Choi, Rafael M. Fernandes, Turan Birol, Liuyan Zhao, Jong Seok Lee, Bharat Jalan
Proc. Natl. Acad. Sci. **123** (2026)[Journal] [arXiv]
- Noncollinear Magnetic Multipoles in Collinear Altermagnets
Luca Buiarelli, Rafael M. Fernandes, Turan Birol
Phys. Rev. B **112**, 224442 (2025) [Journal] [arXiv]

Preprints

- Piezomagnetism in a model cubic noncollinear altermagnet
Sudarshan Sharma, **Luca Buiarelli**, Richard Spieker, Ivan Jakovac, Damjan Pelc, Turan Birol, Martin Greven
Submitted [arXiv]
- Discovery of an odd-parity f-wave charge order in a kagome metal

Jiangchang Zheng, Caiyun Chen, Ruiqin Fu, **Luca Buiarelli**, Zihan Lin, Fazhi Yang, Tianhao Guo, Ganesh Pokharel, Andrea Capa Salinas, Sen Zhou, Turan Birol, Stephen D. Wilson, Junzhang Ma, Daniel J. Schultz, Xianxin Wu, Berthold Jäck
Submitted [arXiv]

In preparation

1. Nonrelativistic spin-lattice coupling in g -wave altermagnets: signatures in diffraction, Raman, and elastic constants

Luca Buiarelli, Rafael M. Fernandes, Turan Birol

2. Interplay of spin-orbit coupling, occupation, and atomic displacements in altermagnetic piezomagnetism

Luca Buiarelli, Rafael M. Fernandes, Turan Birol

3. 3/8 instability in the Kagome metal CsV_3Sb_5

Luca Buiarelli, Hyeonseo H. Park, Ethan Ritz, Rafael M. Fernandes, Turan Birol

Workshops and Conferences

- Aspen Center for Physics, Winter Conference on Altermagnetism, Aspen *April 2026*
“Altermagnetic piezomagnetism and spin-lattice coupling” (poster)
- American Physical Society March Meeting, Denver *March 2026*
“ProDenCeR: a python code to project first-principles densities onto harmonics and representations” (contributed talk)
- Flatiron Institute, TRIQS School, Paris *September 2025*
- American Physical Society March Meeting, Los Angeles *March 2025*
“Magnetic multipoles in altermagnets” (contributed talk)
- School on Electron-Phonon Physics, University of Texas at Austin *June 2024*
- American Physical Society March Meeting, Minneapolis *March 2024*
“Structural properties of kagome-layered crystals” (contributed talk)

Awards and Fellowships

Matthew Tirrell Graduate Fellowship *2024*

Teaching

Teaching Assistant *Spring and Fall 2024*
University of Minnesota, Introduction to Materials Science Laboratory (MATS 2001/2002)

Service and Outreach

Journal Refereeing

Nature *npj Quantum Materials*, Nature *Communications Materials*

Penne Amiche della Scienza

2025-Present

Science outreach program for Italian middle school students involving monthly letter exchanges, aimed at fostering interest in science and academic careers.

Skills

Coding: Python, Matlab, C/C++

First-principles: Abinit, VASP, Wannier90, pythtb

Scientific Software Development: Developed multiple post-processing tools for DFT and Wannier tight-binding Hamiltonians, including ProDenCeR and an independent implementation of Wannier-based nonlinear optical conductivity interpolation.

Languages: English (fluent), Italian (native), Spanish (basic)